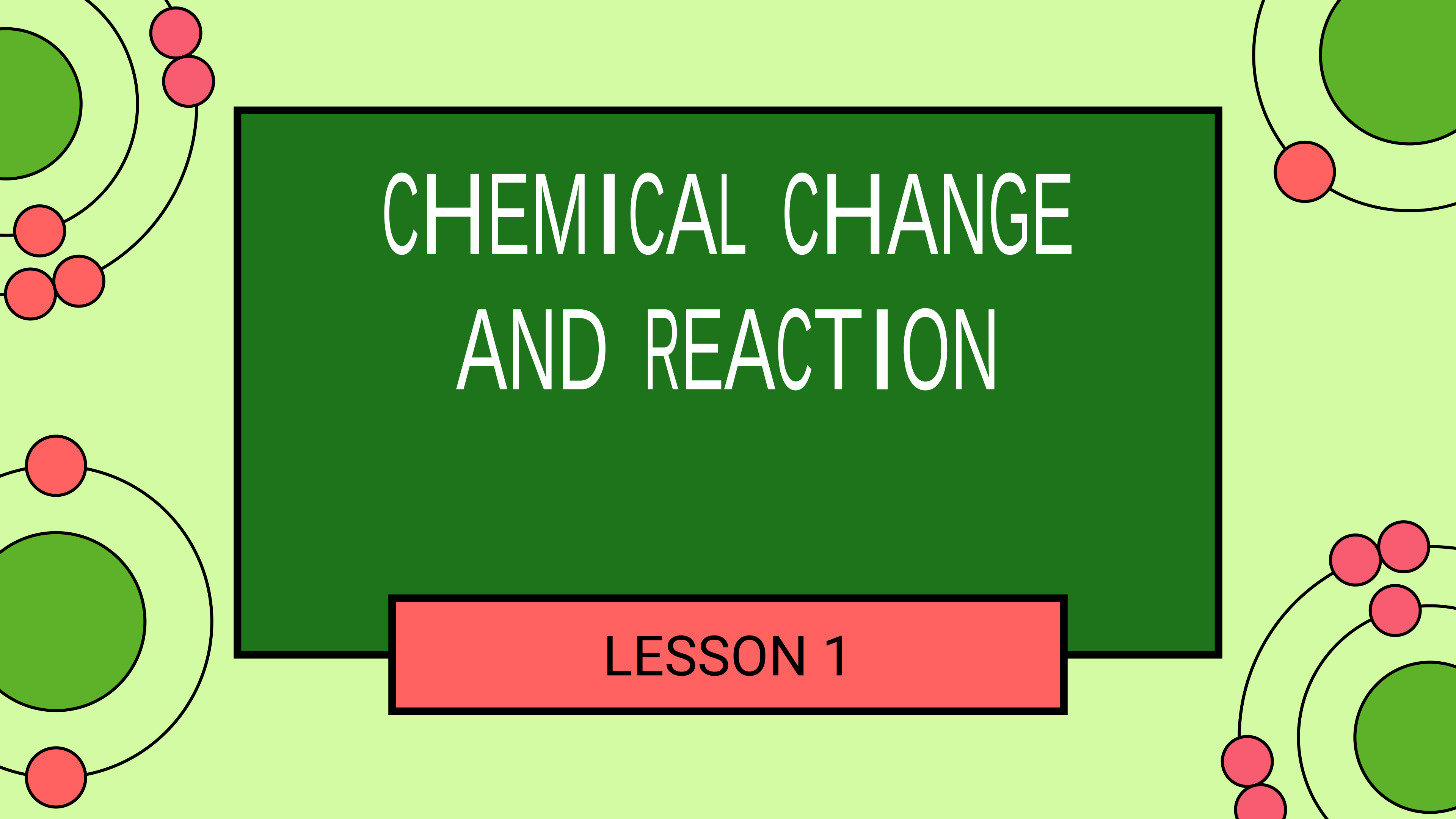
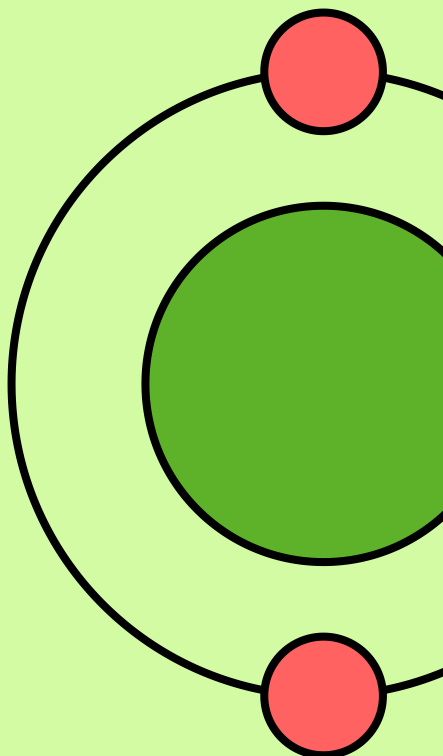
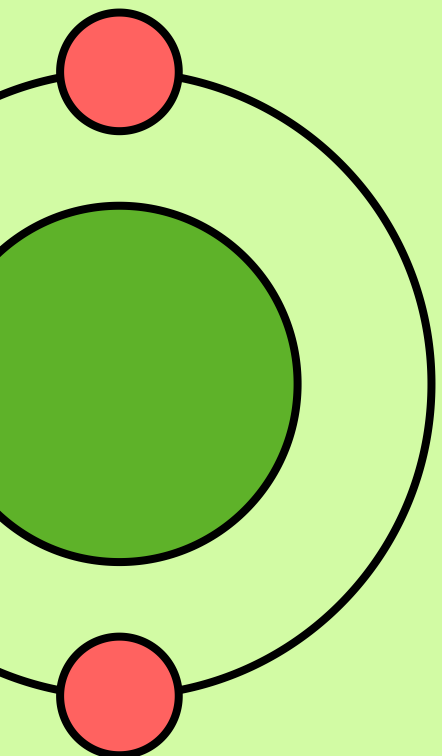


The background is a light green color. In each of the four corners, there is a stylized atomic model. Each model consists of a large green circle representing the nucleus, surrounded by a black circular line representing the electron shell. Small red circles representing electrons are placed on these lines. Top-left: three red electrons. Top-right: one red electron. Bottom-left: one red electron. Bottom-right: three red electrons.

CHEMICAL CHANGE AND REACTION

LESSON 1



LEARNING POINTS

1

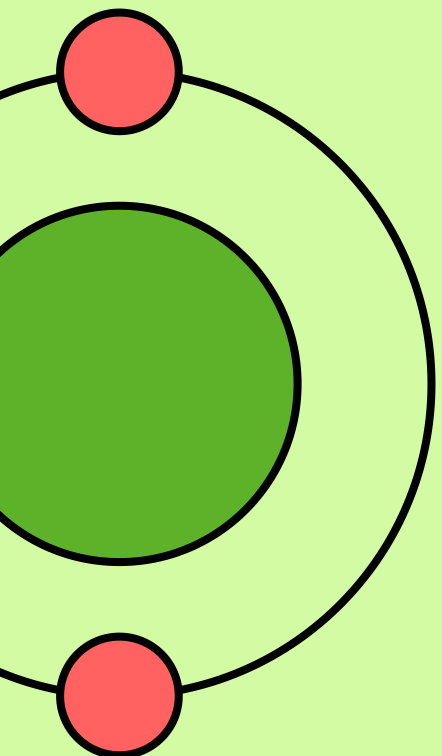
describe how a
chemical reaction
takes place

2

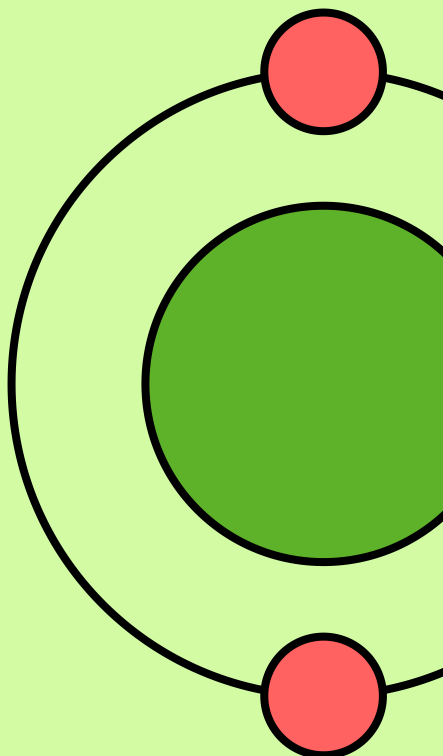
explain the occurrence
of chemical change
and chemical reaction

3

differentiate
exothermic and
endothermic



LEARNING POINTS



4

describe the different indicators of chemical reaction

5

enumerate the indicators of chemical reactions.

WHAT HAPPENS WHEN POPCORN POPS?

When exposed to heat, popcorn kernels transform into fluffy popcorn – an example of a chemical change.



- Chemical changes and reactions are processes where substances turn into new substances with different properties.

IMPORTANCE OF CHEMICAL REACTIONS

- Powering vehicles through combustion
- Producing medicines and materials

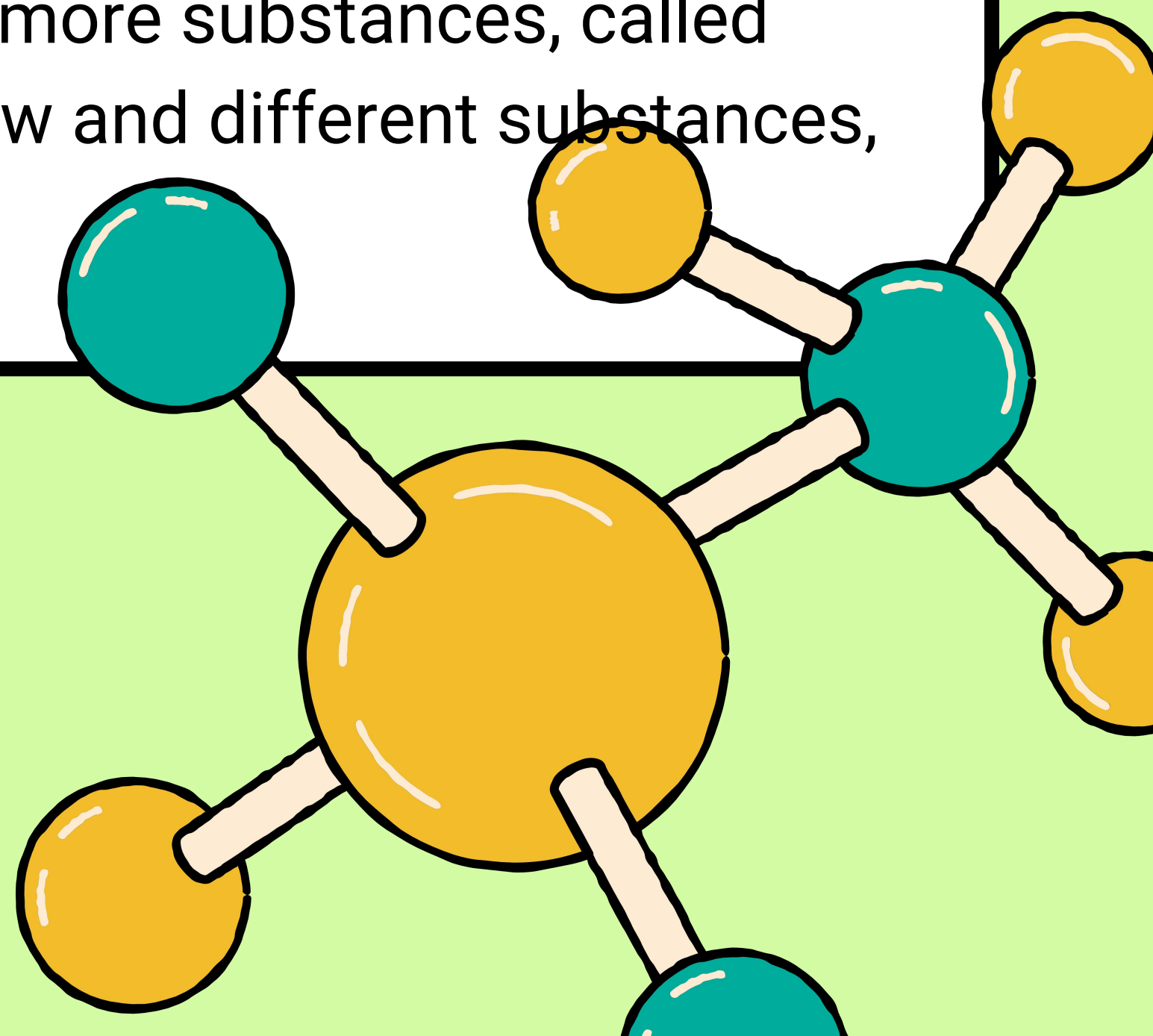


- Generating electricity in power plants
- Cooking and baking processes

WHAT IS A CHEMICAL REACTION?

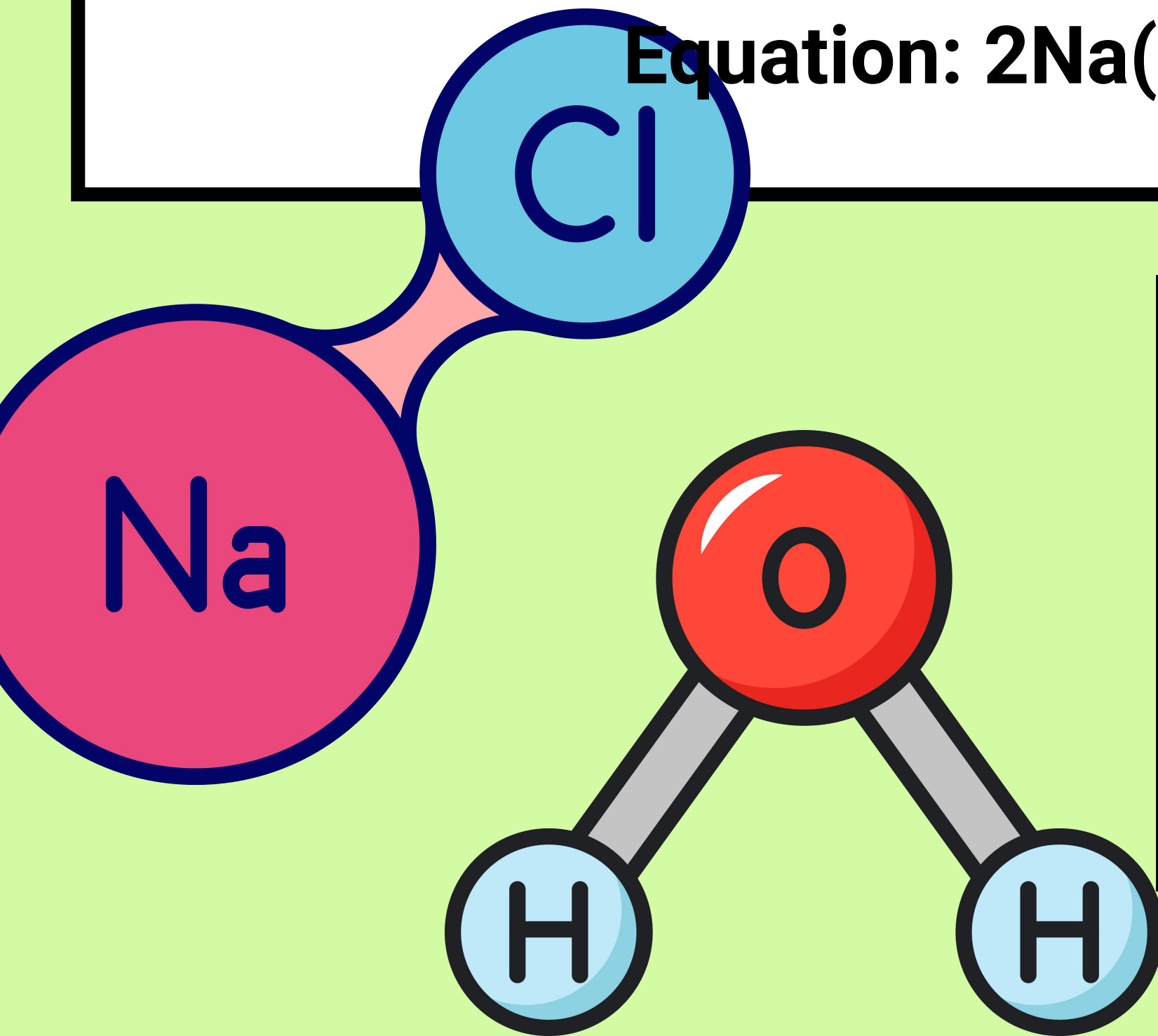
A chemical reaction is a process where one or more substances, called reactants, are transformed into one or more new and different substances, called products.

- Reactants: Starting materials
- Products: New substances formed
- Represented by a balanced chemical equation:
- Example: Reactants $\rightarrow$ Products



EXAMPLE: FORMATION OF SODIUM CHLORIDE

Sodium (silver metal) + Chlorine (yellow gas) → Sodium chloride (white solid)

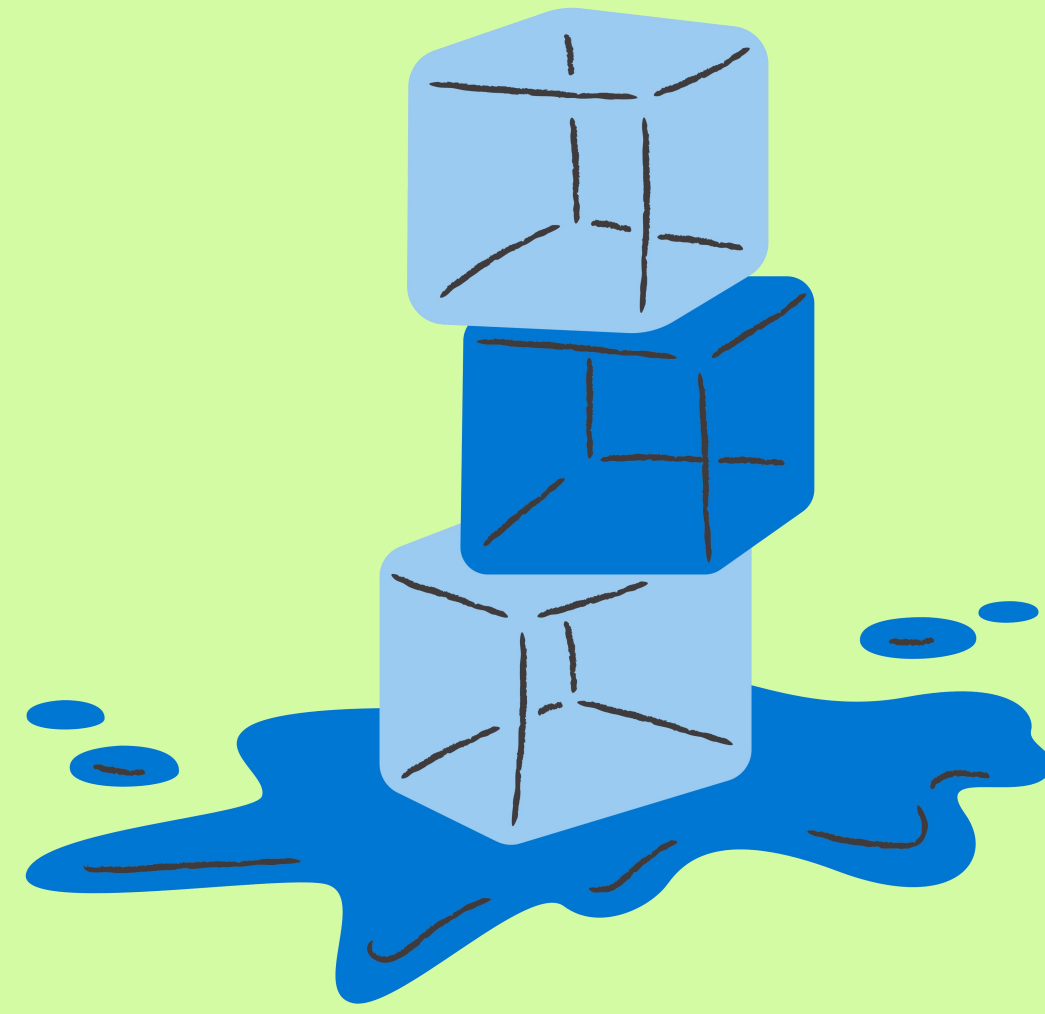
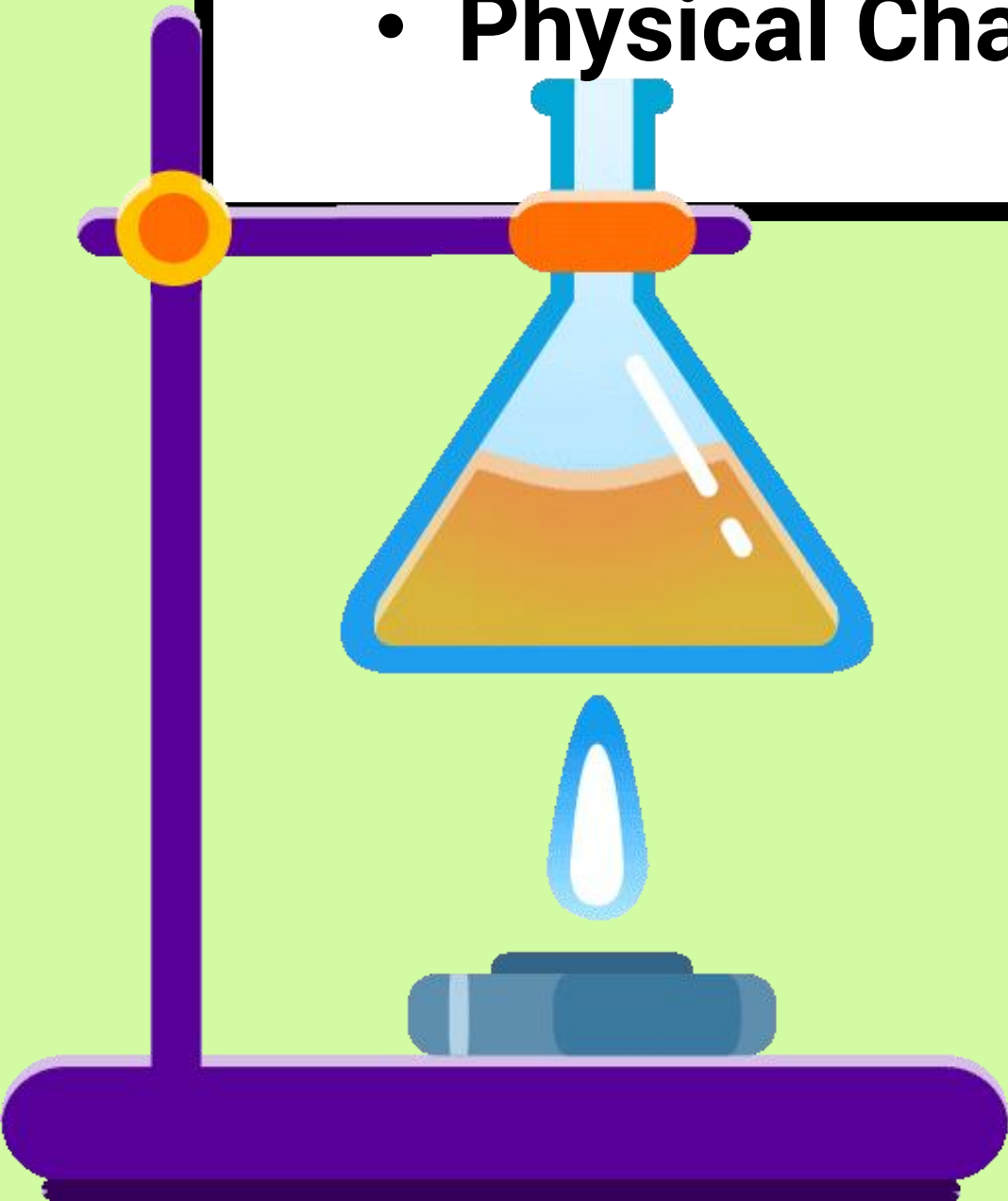


- **Key Points:**

1. **Atoms rearrange, but total atoms stay the same.**
2. **Original substances lose identity.**
3. **Energy is absorbed or released.**
4. **Total mass of reactants = total mass of products**

CHEMICAL VS. PHYSICAL CHANGE

- **Chemical Change:** Produces new substances (e.g., burning wood)
- **Physical Change:** No new substances (e.g., melting ice → water)



INDICATORS OF CHEMICAL REACTIONS

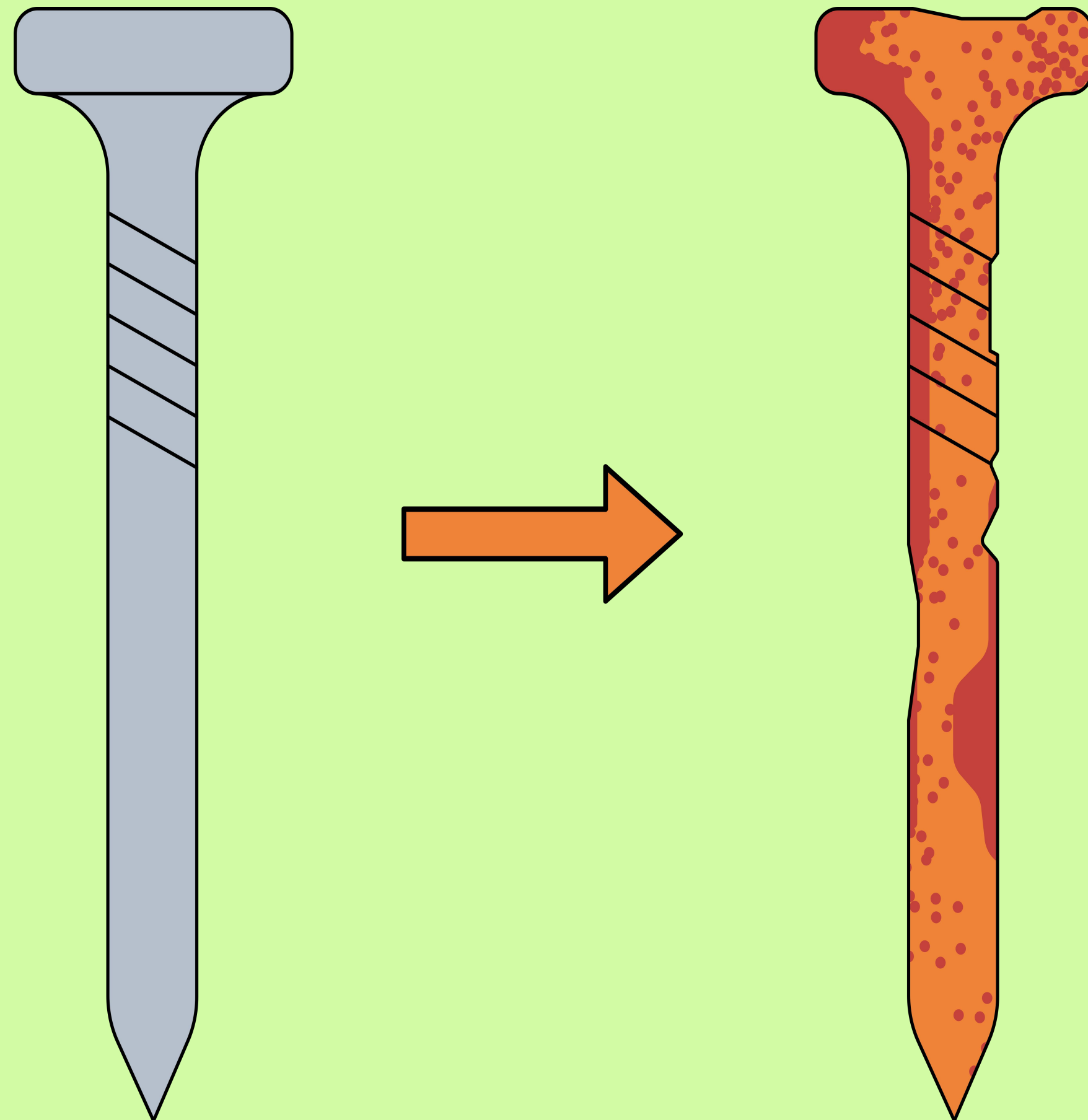
COMMON INDICATORS:

1. Color Change
2. Formation of Precipitate
3. Gas Production
4. Temperature Change
5. Light Emission
6. Odor Change
7. Change in Taste
8. Formation of New Substances

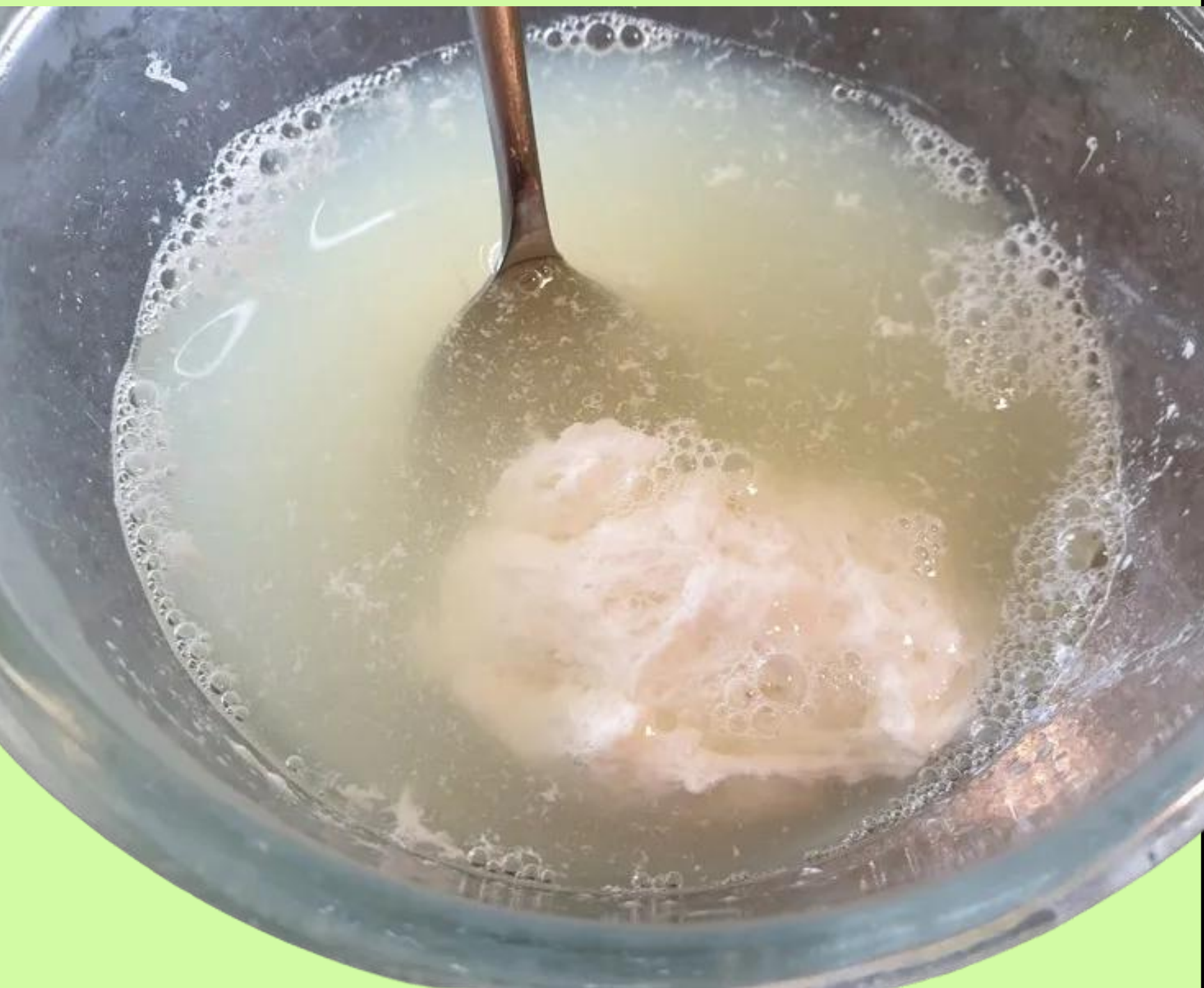
Indicators of chemical reactions are observable clues or changes that suggest that a chemical change is taking place. These indicators help us identify and recognize that a chemical reaction is occurring.

COLOR CHANGE

A change in color often signals that a new substance has been formed. This happens because atoms rearrange to create new compounds with different optical properties. The original material's color fades or changes, indicating that a chemical transformation has taken place rather than a simple physical alteration.



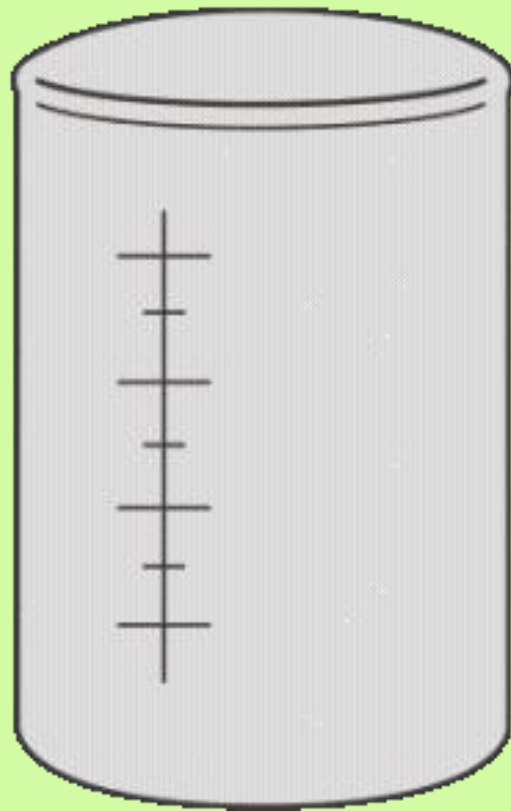
FORMATION OF PRECIPITATE



A precipitate forms when two clear solutions react to produce an insoluble solid that settles out or appears cloudy. .

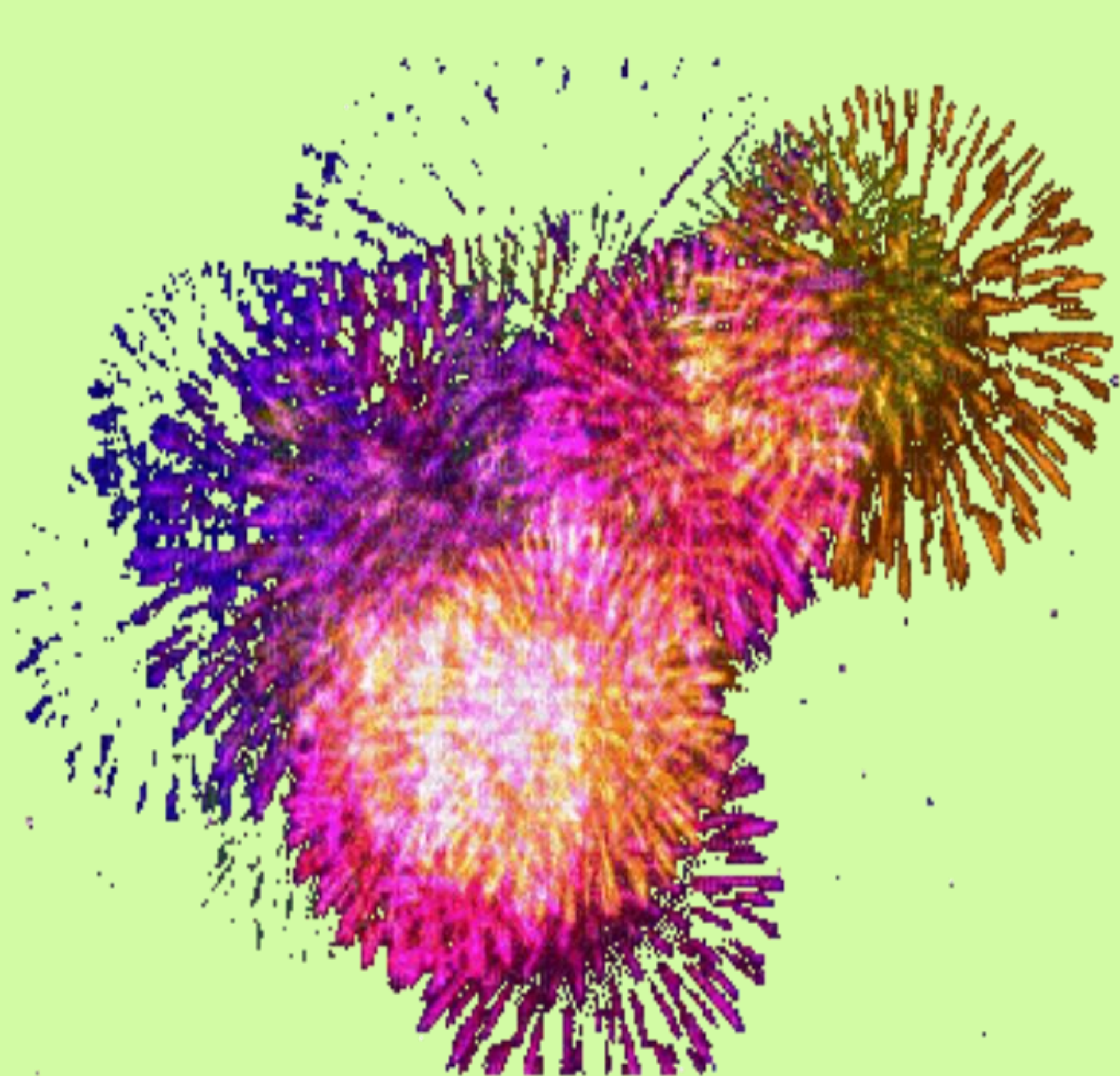
GAS PRODUCTION

The release of gas bubbles during a reaction shows that new gaseous products are being generated. This is an indication that the reactant molecules are breaking apart and recombining in new ways.



LIGHT EMISSION

The production of light during a reaction, such as in fireworks or glow sticks, means that energy is released in the form of light as atoms and molecules rearrange.



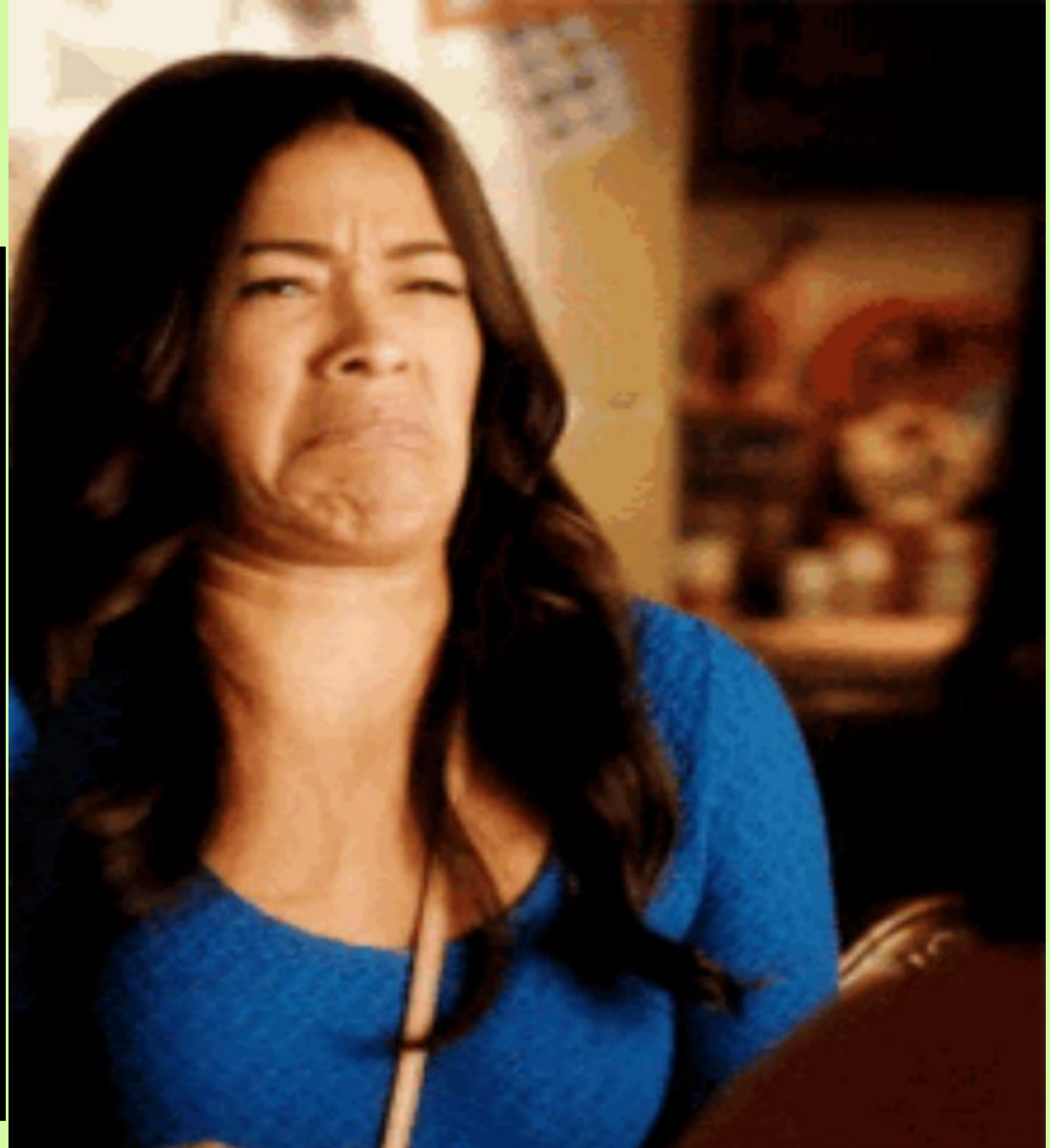
ODOR CHANGE

A new or altered smell suggests that new volatile compounds have been produced. Since odor is linked to the molecular structure of substances, a change in smell confirms that new chemical compounds with different structures have been formed.



CHANGE IN TASTE

A shift in taste also indicates that a new chemical substance has been formed, as taste depends on the molecular composition of compounds. When chemical reactions occur—especially in cooking or food preparation—the original flavors change due to the formation of new flavor compounds.



FORMATION OF NEW SUBSTANCES

The most conclusive indicator of a chemical reaction is the creation of entirely new substances with new properties. When the identity, structure, and behavior of the original materials change completely, it signifies that a chemical change has taken place.



Energy in Chemical Reactions

- Chemical reactions involve energy changes.
- Based on how they absorb or release energy, they are classified as endothermic or exothermic.
- All chemical reactions involve breaking and forming of bonds.
These processes either absorb or release energy, which affects the temperature of the surroundings. That's why some reactions feel warm while others feel cold.

ENDOTHERMIC REACTIONS

In an endothermic reaction, the system needs to absorb energy to break existing chemical bonds. This energy is usually in the form of heat, which is drawn from the surroundings, making the environment feel cooler. The total energy of the products becomes higher than that of the reactants because of the absorbed energy.

- Absorb heat or energy from surroundings.
- Cause the temperature of surroundings to decrease.
- Reactants have less energy than products.

CHARACTERISTICS OF ENDOTHERMIC REACTIONS

1. Heat Absorption – Energy enters the system from surroundings.
2. Temperature Decrease – The surroundings become cooler.

Examples:

- Melting of ice
- Dissolving ammonium nitrate in water
- Photosynthesis
- Cooking an egg



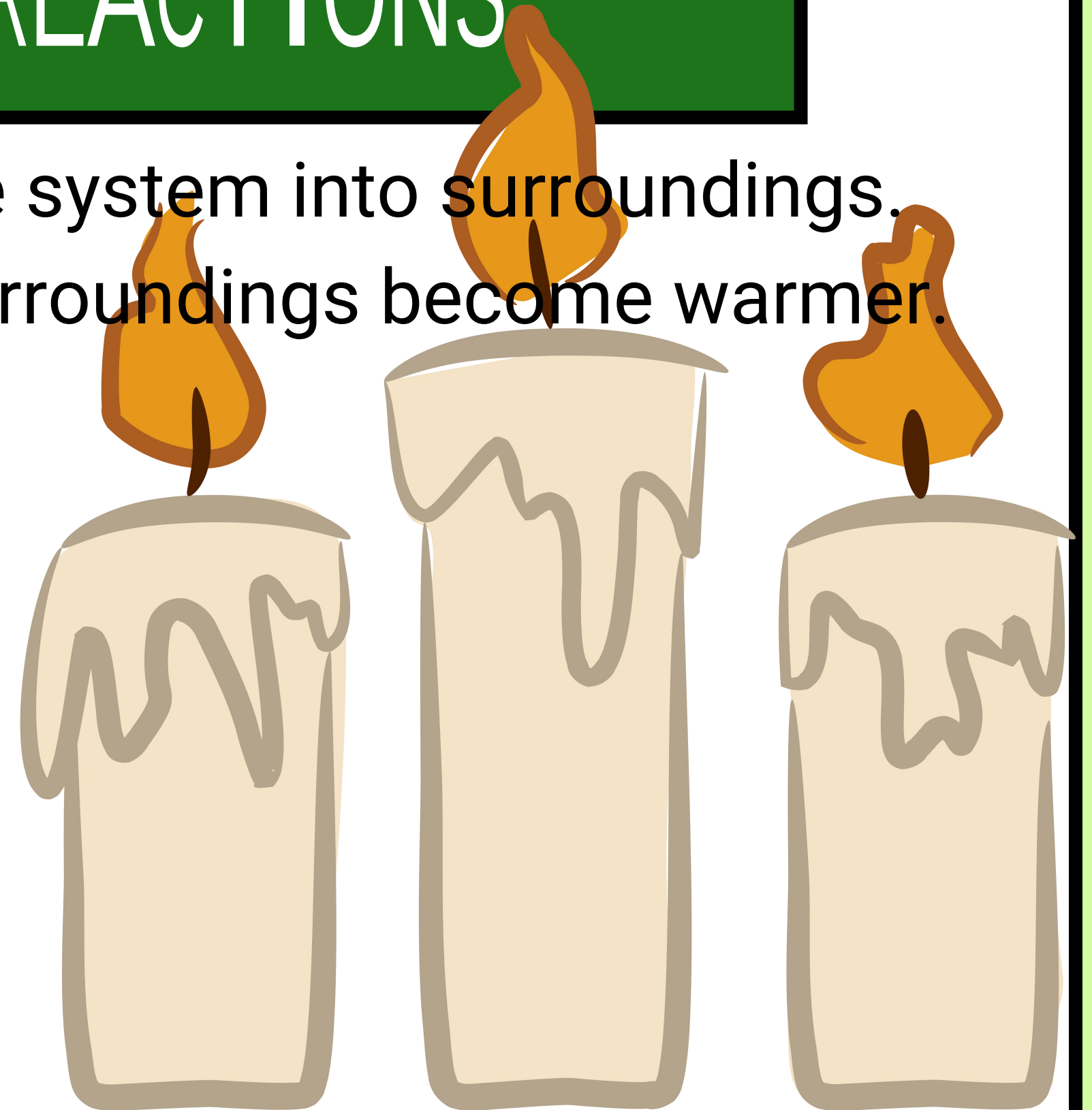
EXOTHERMIC REACTIONS

In an exothermic reaction, energy is released as new bonds form. This release of energy, mostly as heat or light, causes the temperature of the surroundings to rise. Because the system loses energy, the products have lower energy than the reactants.

- Release heat or energy to surroundings.
- Cause the temperature of surroundings to increase.
- Reactants have more energy than products.

CHARACTERISTICS OF EXOTHERMIC REACTIONS

1. Heat Release – Energy exits the system into surroundings.
2. Temperature Increase – The surroundings become warmer.
3. Examples:
4. Combustion of fuels
5. Rusting of iron
6. Reaction of acid and base
7. Lighting fireworks or candles



EXOTHERMIC VS. ENDOTHERMIC REACTIONS

Exothermic Reactions	Endothermic Reactions
Release heat energy	Absorb heat energy
Surroundings become warmer	Surroundings become cooler
Reactants have higher energy	Products have higher energy
Example: Combustion, Rusting, Fireworks	Example: Photosynthesis, Ice melting, Cooking an egg

The background is a light green color. It features decorative elements in the corners: large green circles and smaller pink circles connected by thin black lines. These elements are positioned in the top-left, top-right, bottom-left, and bottom-right corners, framing the central text area.

THANK YOU FOR LISTENING!